

Anamika Saha

Software Engineer • Full-Stack Developer • Applied AI/ML • BI & Data Operations

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PROFILE

Computer Science graduate with research interests in Artificial Intelligence, Machine Learning, Data Science, Deep Learning, Natural Language Processing and Health Informatics. Experienced in Full-Stack Development, Business Intelligence and Data Operations, with hands-on expertise in Python, JavaScript, React.js, Node.js, Oracle SQL, ETL processes and enterprise database management. Currently working as a Junior Software Engineer while actively contributing to AI research through peer-reviewed publications and ongoing research projects.

EDUCATION

Ahsanullah University of Science & Technology

B.Sc. in Computer Science & Engineering (CGPA: 2.87/4.00 | Last Two Years CGPA: 3.06/4.00)

2021 – 2025

Viqarunnisa Noon College

Higher Secondary Certificate (Science) (GPA: 5.00/5.00)

2018 – 2020

Siddheswari Girls' High School

Secondary School Certificate (Science) (GPA: 5.00/5.00)

2016 – 2018

- Government Scholarship Recipient

PROFESSIONAL EXPERIENCE

• Jr. Software Engineer (Business Intelligence & Data Maintenance)

Jan 2026 – Present

Teletalk Bangladesh Ltd. (through Opus Technology Limited)

Gulshan, Dhaka

- Managed **Tableau Server monitoring** and **BI reporting** workflows to support telecom data-driven decision making.
- Worked with **Oracle databases** for tablespace monitoring, **SQL validation** and data loading using SQL Developer.
- Gained working knowledge of **Data Warehouse architecture** and **ETL pipeline** processes in telecom BI systems.
- Performed in telecom data analysis, including **CDR log monitoring** and basic CDR validation.
- Automated server maintenance tasks through **cron jobs** and **bash scripting**, optimizing pipeline flow and system monitoring.

PROJECTS

• Teletalk Subscriber Analytics Bot | Python, FastAPI, Llama 3.2, Oracle, ML

View Project

Built an AI-powered subscriber analytics chatbot for Teletalk Bangladesh featuring natural language querying and ML-based subscriber clustering.

• CDR14 Web Application | Python, Flask, Oracle SQL, PyQt5, HTML

View Project

Enhanced Teletalk's internal CDR investigation system with time-based filtering, Oracle SQL integration, a Flask web interface and CSV export.

• CDR Log Monitor Dashboard | Python, Flask, Paramiko, Multithreading, REST API

View Project

Built a real-time CDR pipeline monitoring dashboard for Teletalk tracking file registration, merge, and L1 loading stages across multiple CBS and MSC servers via SSH with live status updates.

• Mystro - The Fighting Game | C++, iGraphics, Visual Studio IDE

View Project

Developed a 2D fighting game implementing game logic, interactive graphics and real-time rendering, demonstrating strong OOP principles.

• Foodie | C#, ASP.NET MVC, MySQL, Visual Studio

View Project

Developed an online food ordering website featuring interactive menus, order management & responsive design.

• Interactive 3D Book Experience | Three.js, WebGL, Shaders

View Project

Developed a web-based immersive 3D book with realistic page-turning physics, custom shaders and 360° controls.

SKILLS

• Hard Skills:

Languages: HTML5, CSS3, JavaScript, TypeScript, Python, C, C++, MATLAB, Java(Basic), PHP, SQL

Frontend & Web: React.js, Next.js, Tailwind CSS, Bootstrap, Redux Toolkit, Material-UI

Backend & Cloud: Node.js, Express.js, BashScript, ASP.NET MVC, Flask, FastAPI, REST APIs, Uvicorn, Linux, AWS EC2

Databases: MySQL, MongoDB Atlas, Oracle SQL, Microsoft SQL Server, Firebase

Data & ML Pandas, NumPy, Matplotlib, Seaborn, TensorFlow, PyTorch, NLP

Tools & Design: Tableau, Git, Docker, Netlify, Vercel, Figma, Canva

• Soft Skills:

Problem-Solving & Critical Thinking, Teamwork & Leadership, Technical Communication & Presentation, Time Management & Adaptability, Creativity & Project Management

RESEARCH & PUBLICATIONS

- **Cardiovascular Disease Prediction: A Machine Learning Approach** View Thesis
Thesis Defended
Description: Researched predictive ML models using demographic, clinical and lifestyle data. Evaluated multiple algorithms including **LightGBM, CatBoost, XGBoost and ANN**, achieving **90%+ accuracy**.
- **Common Reasons for Chest X-ray Examinations in Hospitals** View Journal
Journal Published; European Journal of Medical and Health Sciences
Description: Investigated clinical indications, diagnostic outcomes and public awareness of chest X-ray examinations across multiple hospitals in Dhaka to improve diagnostic practices and healthcare awareness.
- **Artificial Intelligence in Chest Radiography Comparative Study with Hospital Radiologists Reports** View Journal
Journal Published; European Journal of Artificial Intelligence and Machine Learning
Description: Evaluated the effectiveness of a DenseNet-121 deep learning model against hospital radiologists' reports, highlighting AI's role in enhancing diagnostic accuracy and supporting clinical workflows.
- **A Deployment and Resource Aware Edge AI Framework for Real-Time Waste Classification with Biometric Authentication and Secure Logging** View Work
Manuscript under review; PLOS ONE; submitted April 2026
Description: Engineered an automated waste management system integrating **Edge-AI** for real-time classification, **Biometric Authentication** and a **Blockchain-based incentive mechanism** to ensure data integrity and transparency.
- **ZETA: A Zero-Knowledge and Threshold-Cryptographic Architecture for Scalable and MEV-Resilient Sharded Blockchains** Manuscript
Manuscript under review; Engineering Reports; submitted July 2026
Description: Proposed a scalable blockchain architecture integrating Zero-Knowledge Proofs, Threshold Cryptography and sharding techniques to improve security, transaction throughput and resistance against MEV attacks.

TRAINING / CERTIFICATIONS

- **Automation Essentials: Industrial Process Control System Design (AUSTRC)** 30 – 05 – 2024
- **Hands-on Workshop on FPGA Acceleration for Machine Learning Applications** 30 – 01 – 2025
- **Workshop on GitHub, CNCF and AWS – AUST Programming and Informatics Club** 26 – 08 – 2025
- **Space Webinar on “Charting Saturn’s Secrets: The Cassini Mission & Beyond”** 23 – 01 – 2025

EXTRA-CURRICULAR ACTIVITIES

- **WebMaster** July 2025 – Dec 2025
AIAA AUST Student Branch
Maintained the branch's official website and technical infrastructure.
Hosted and anchored various online and offline programs and meetings on behalf of the branch.
- **Deputy Executive of GFX Team** May 2024 – Aug 2025
AUST Robotics Club
Led the design team to create event posters and digital content; **Co-led** Guests and Judges Handling for **ARC 2.0** (AUST Rover Challenge).
Served as an organizer, host and anchor for the National Robotics Event's Prize-Giving Ceremony.
- **Graphic Designer & Project Member** 2024 – 2025
Satellite Communication Laboratory, AUST
Designed posters, infographics and visuals for internal and external lab communication.
Worked as a **Logistics sub-team member** for the AUST Nanosatellite Project.
- **Participant, Volunteer & Organizer** Spring 2022 – Spring 2024
AUST CSE Carnival
Volunteered in CSE Carnival 1.0 to 3.0; Participant in CSE Carnival 2.0 to 4.0.
Organizing Committee Member for CSE Carnival 5.0.
- **Campus Ambassador** Mar 2022 – 2023
ICT Olympiad Bangladesh
Coordinated inter-department student participation in various Olympiad events.
Promoted ICT awareness and successfully drove registration campaigns on campus.
- **Sports & Event Organizer** Spring 2021 – Fall 2024
AUST CSE Department
Represented the department in Football and Cricket tournaments across all semesters, serving as Football Team Captain (Fall 2021).
Organized major departmental events including CSE Cricket Tournament, CSE Picnic and CSE Revelry (CSE Fest) - Spring'24.

REFERENCES

- **Ms. Syeda Shabnam Hasan** – Assistant Professor, Dept. of CSE, AUST – shabnam.cse@aust.edu
- **Mr. Saha Reno** – Assistant Professor, Dept. of CSE, AUST – reno.cse@aust.edu